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(19) **United States**(12) **Patent Application Publication**
KOSHIMIZU(10) **Pub. No.: US 2025/0285839 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **PLASMA PROCESSING APPARATUS****Publication Classification**(71) Applicant: **Tokyo Electron Limited**, Tokyo (JP)(51) **Int. Cl.**
H01J 37/32 (2006.01)(72) Inventor: **Chishio KOSHIMIZU**, Kurokawa-gun (JP)(52) **U.S. Cl.**
CPC .. **H01J 37/32174** (2013.01); **H01J 37/32568** (2013.01); **H01J 37/32642** (2013.01); **H01J 37/32715** (2013.01)(73) Assignee: **Tokyo Electron Limited**, Tokyo (JP)(57) **ABSTRACT**(21) Appl. No.: **19/215,904**(22) Filed: **May 22, 2025****Related U.S. Application Data**

(63) Continuation of application No. 17/190,552, filed on Mar. 3, 2021, now Pat. No. 12,334,305.

(30) **Foreign Application Priority Data**

Mar. 17, 2020 (JP) 2020-046282

A disclosed plasma processing apparatus includes a substrate support. The substrate support has a first region configured to support a substrate and a second region configured to support an edge ring. The first electrode is provided in the first region. The second electrode is provided in the second region. The first bias power source is connected to the first electrode via the first circuit. The second bias power source is connected to the second electrode via the second circuit. The second circuit has impedance higher than impedance of the first circuit at a common bias frequency of a first electrical bias generated by the first bias power source and a second electrical bias generated by the second bias power source.

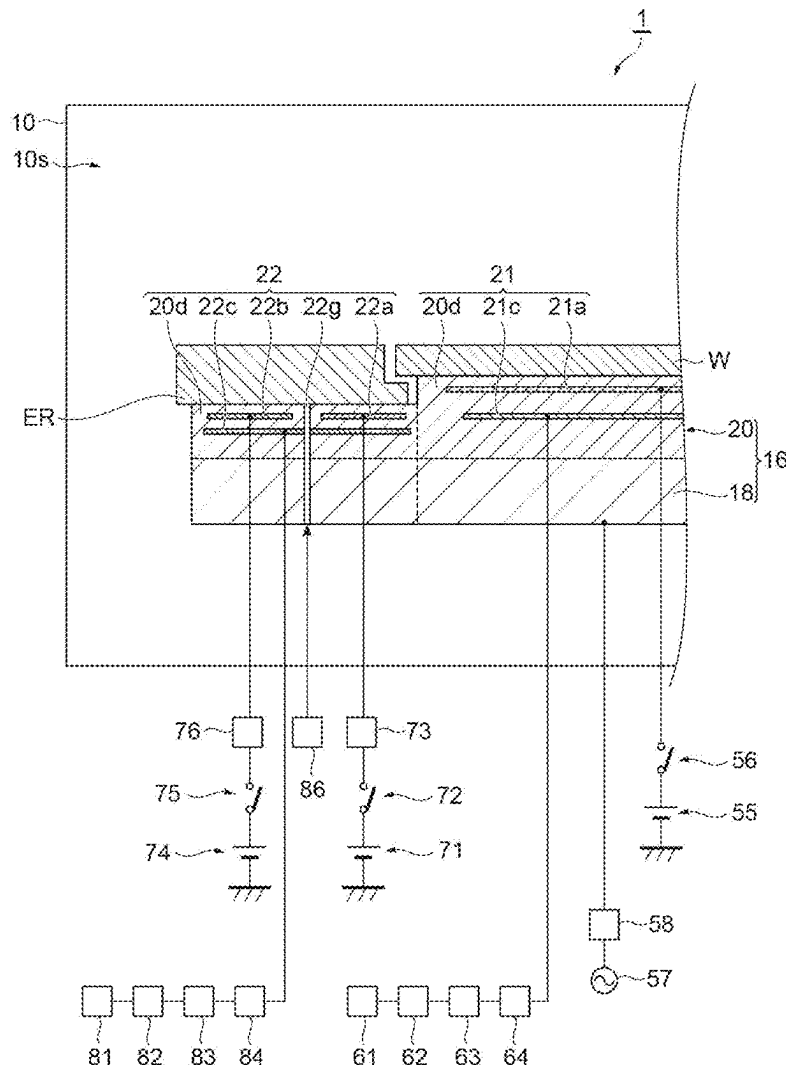


Fig.1

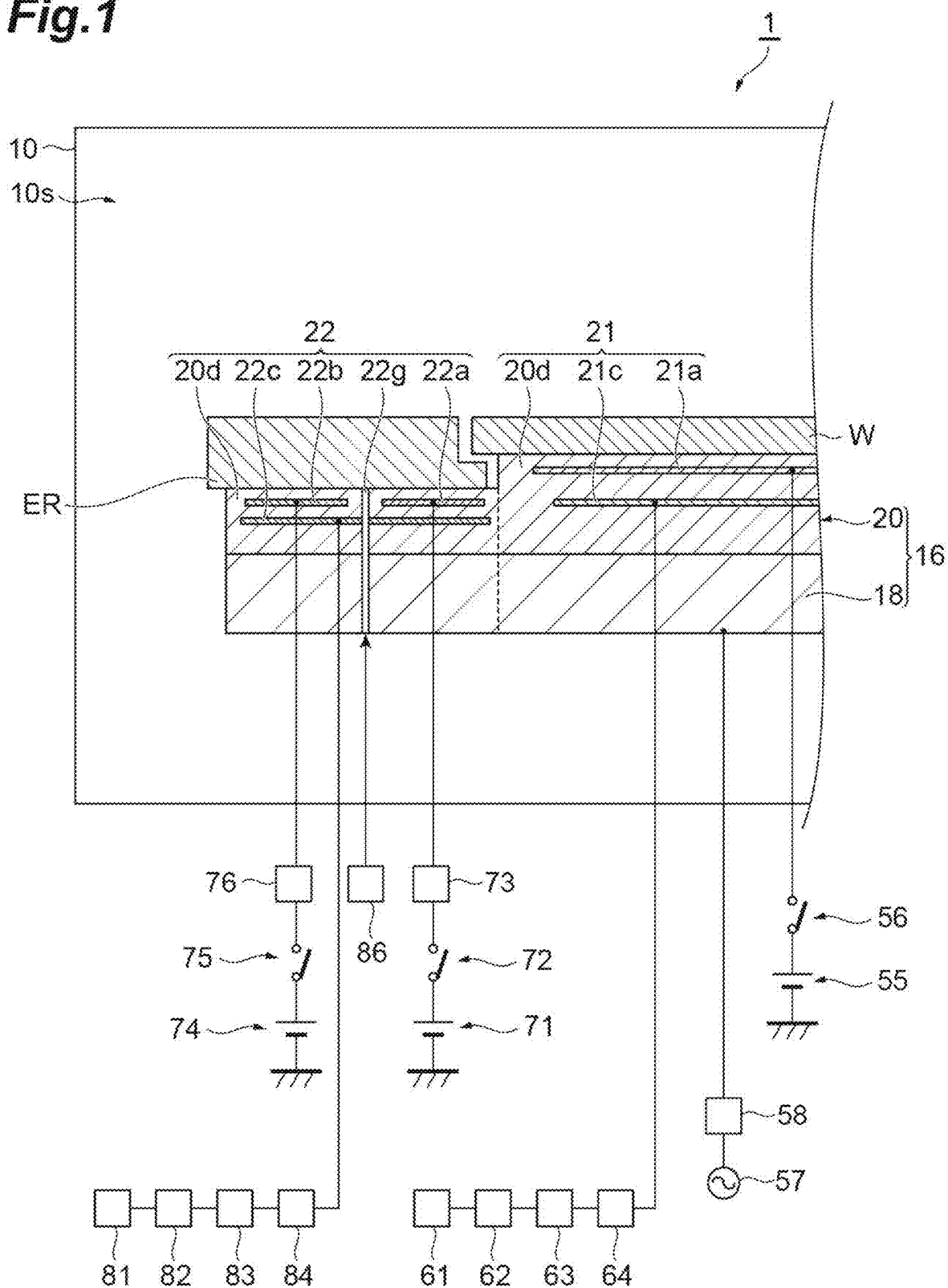


Fig. 2

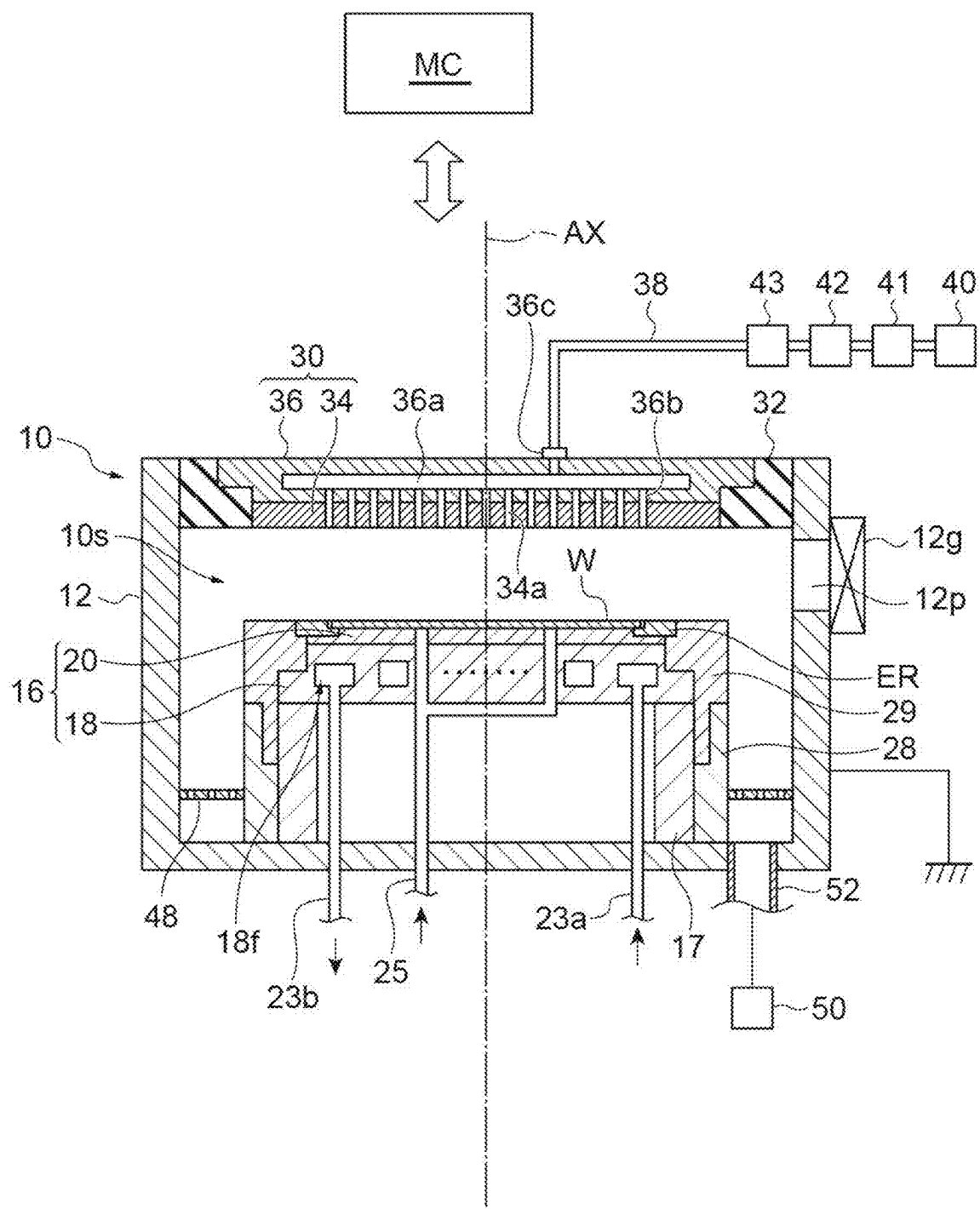


Fig.3

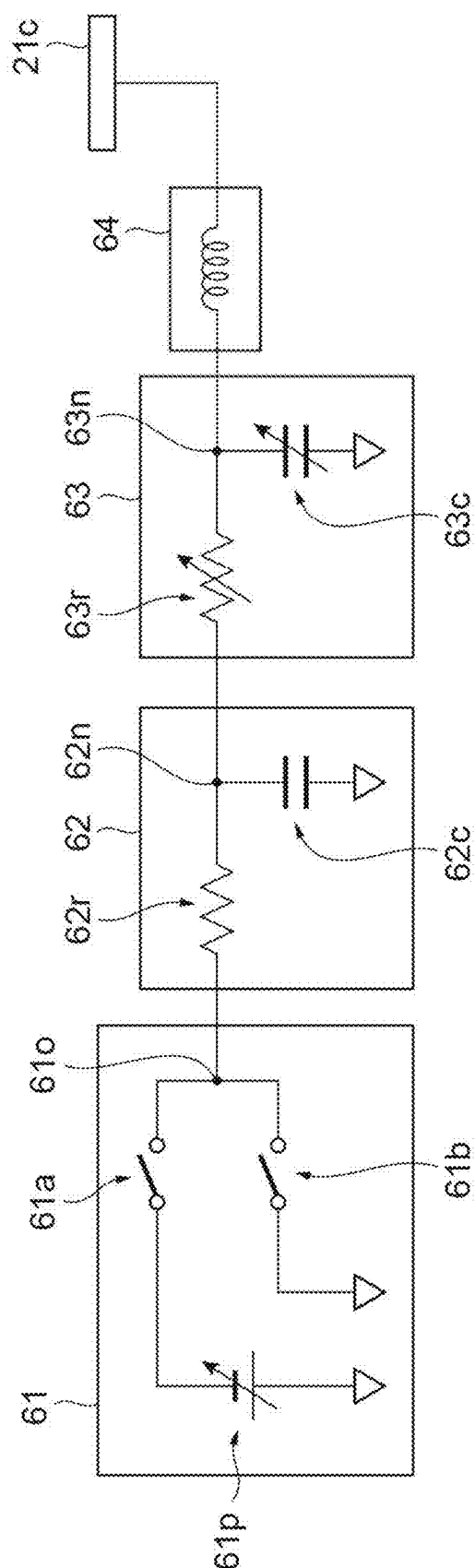


Fig.4

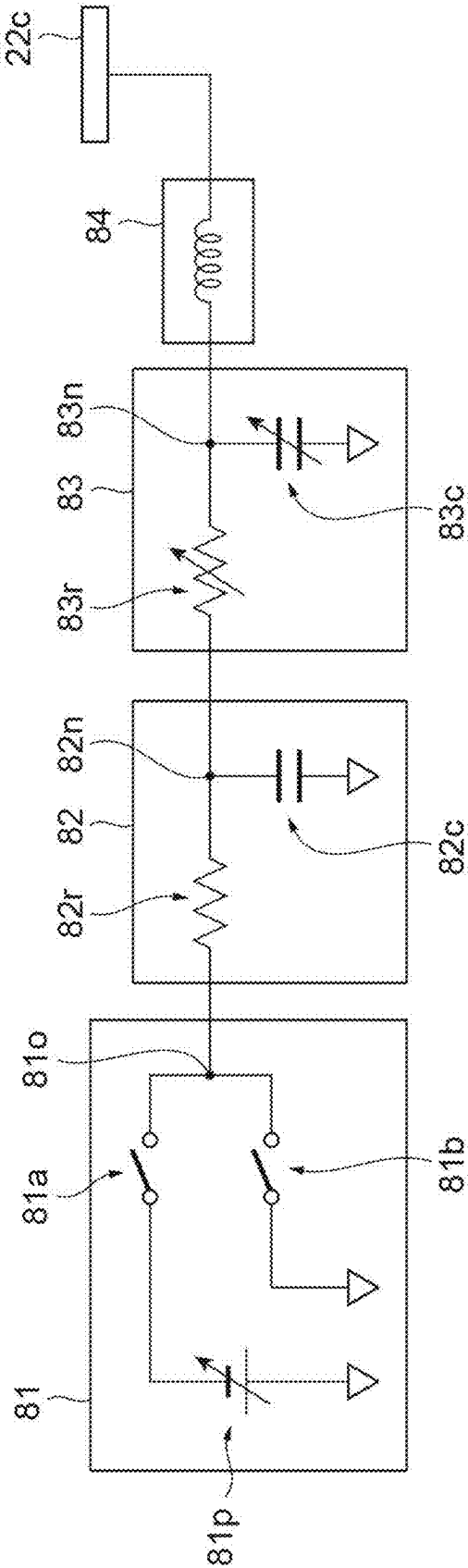


Fig.5A

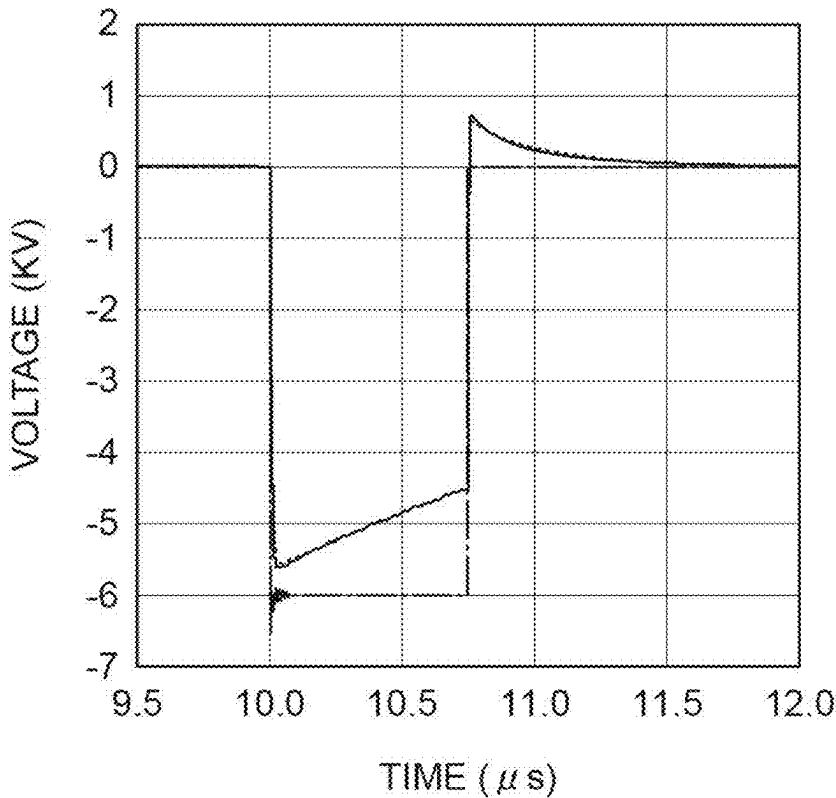


Fig.5B

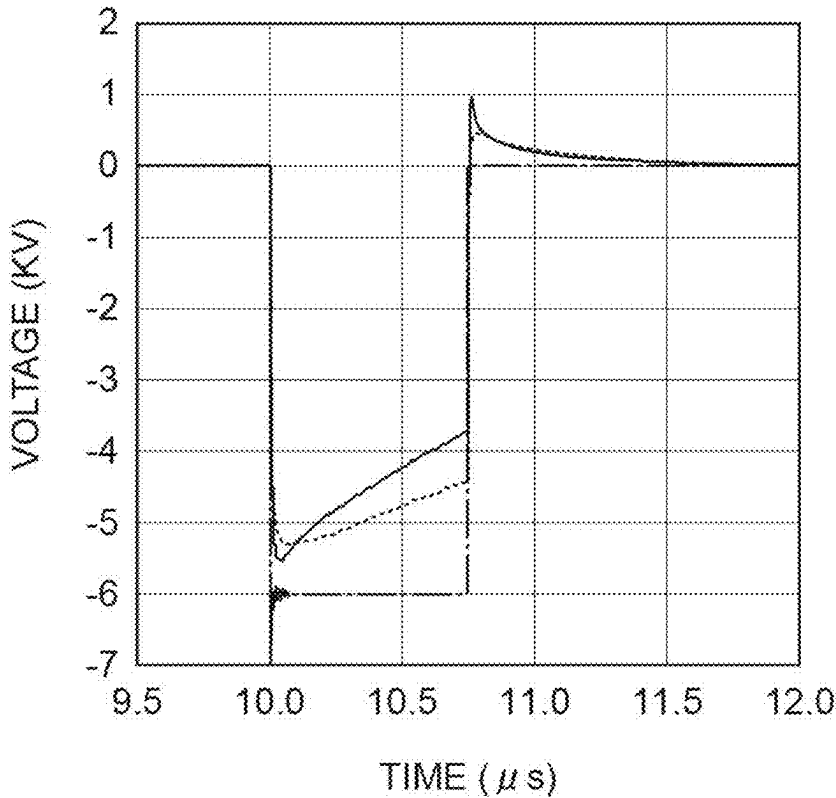


Fig.6

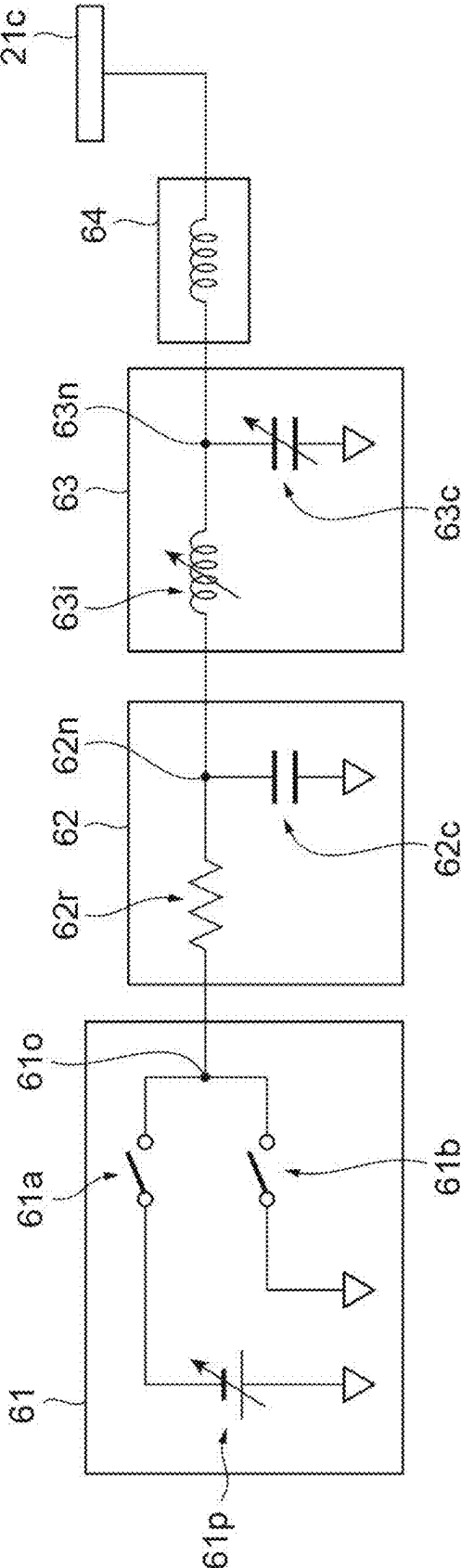


Fig.7

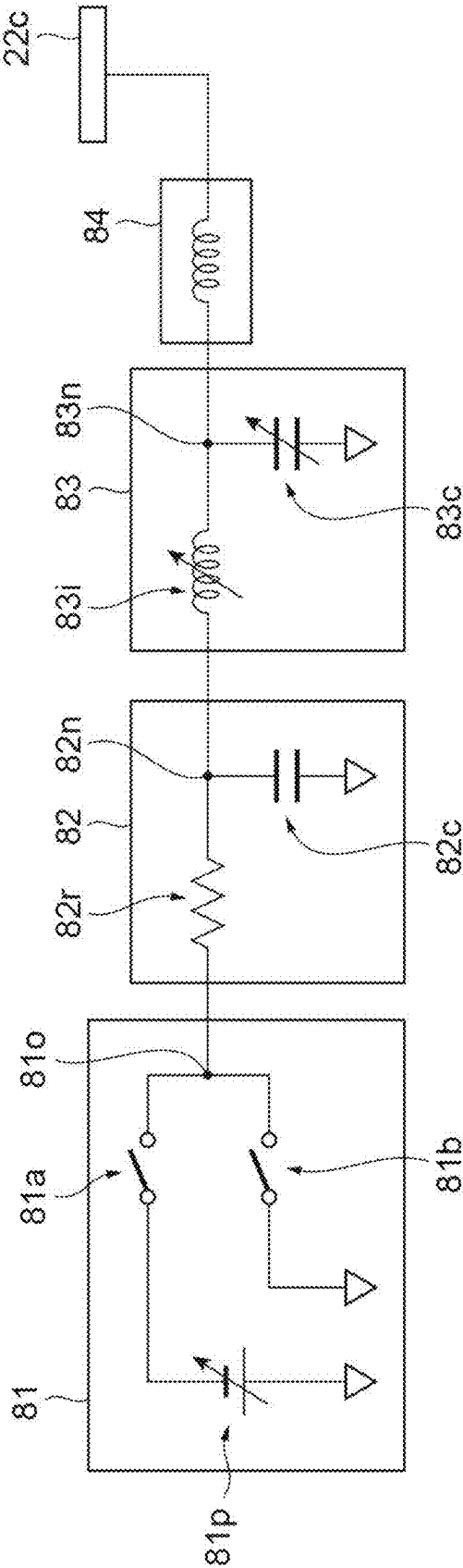


Fig. 8

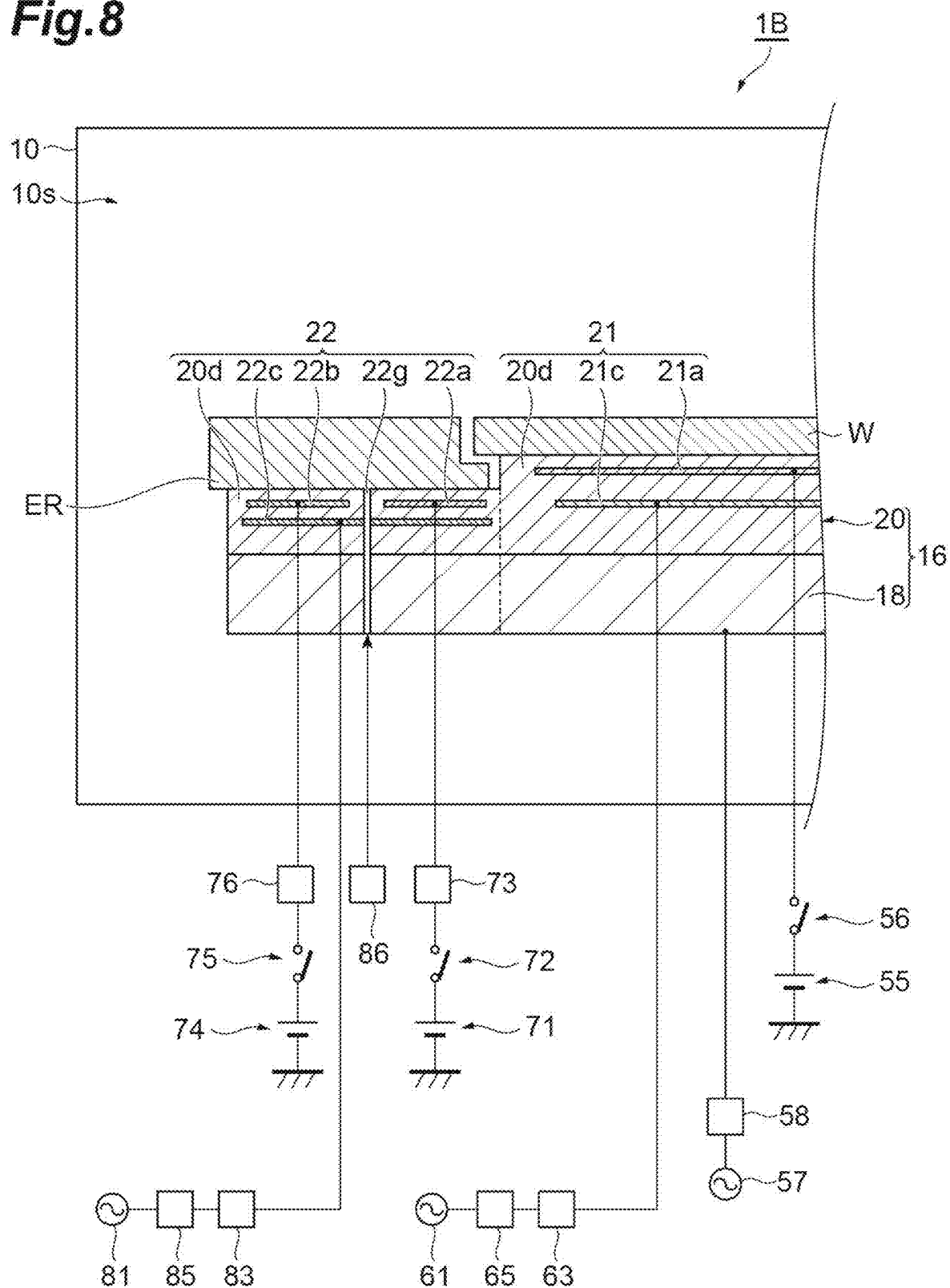
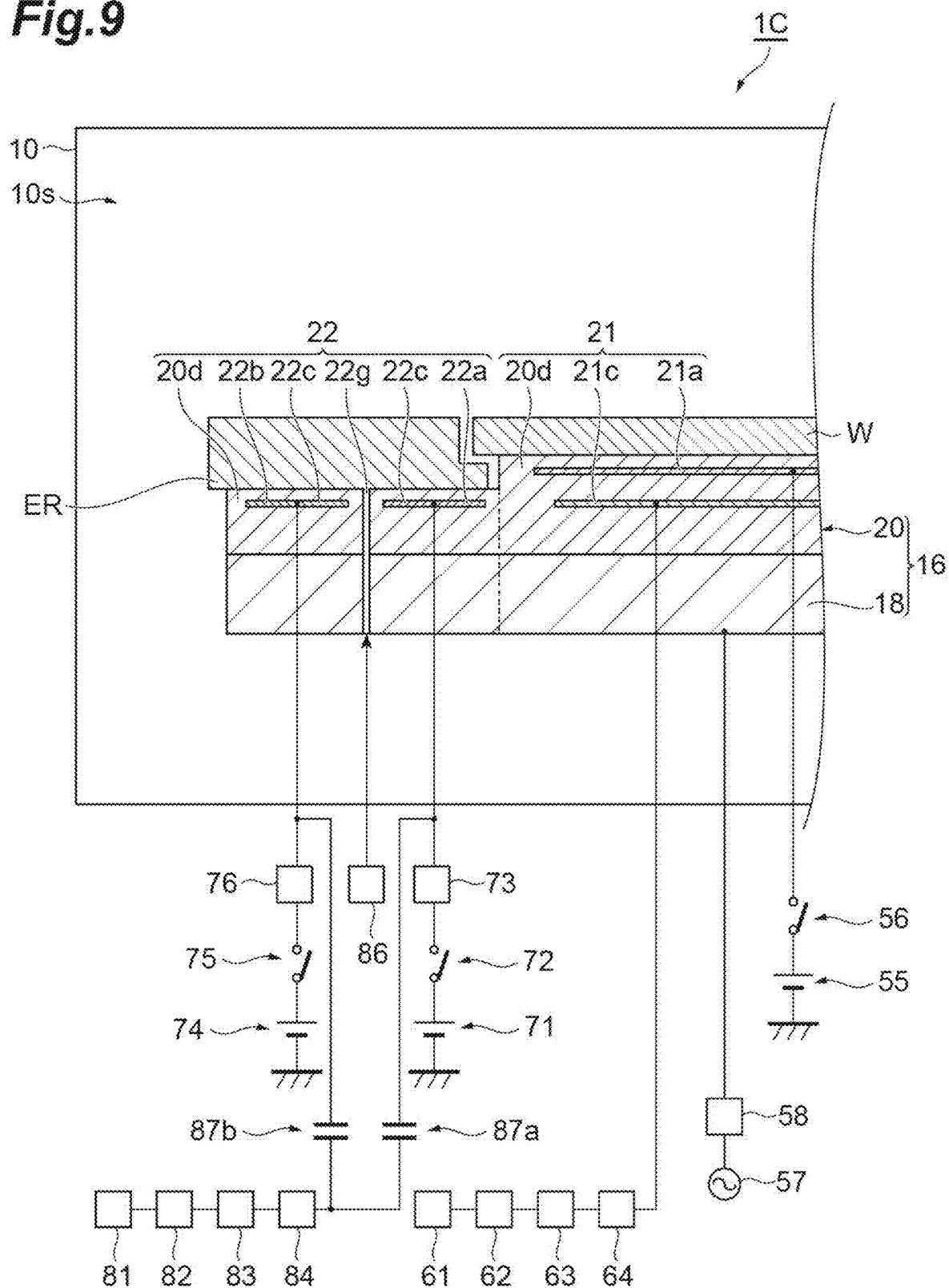


Fig.9



PLASMA PROCESSING APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Continuation of U.S. patent application Ser. No. 17/190,552 filed on Mar. 3, 2021, which is based on and claims the benefit of priority from Japanese Patent Application No. 2020-046282 filed on Mar. 17, 2020, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Exemplary embodiments of the present disclosure relate to a plasma processing apparatus.

BACKGROUND

[0003] In the manufacture of electronic devices, a plasma processing apparatus is used. The plasma processing apparatus has a chamber and a substrate support. The substrate support has a lower electrode and an electrostatic chuck. The electrostatic chuck is provided on the lower electrode. The substrate support supports an edge ring. A substrate is placed in a region surrounded by the edge ring on the substrate support. Radio frequency bias power is supplied to the lower electrode in order to draw ions from plasma into the substrate. Japanese Unexamined Patent Publication No. 2019-36658 discloses such a plasma processing apparatus.

SUMMARY

[0004] In an exemplary embodiment, a plasma processing apparatus is provided. The plasma processing apparatus includes a chamber, a substrate support, a first bias power source, a second bias power source, a first circuit, and a second circuit. The substrate support has a first region, a second region, a first electrode, and a second electrode. The first region is configured to support a substrate. The second region is configured to support an edge ring. The first electrode is provided in the first region. The second electrode is provided in the second region and separated from the first electrode. The first bias power source is configured to generate a first electrical bias and electrically connected to the first electrode. The second bias power source is configured to generate a second electrical bias and electrically connected to the second electrode. The first circuit is connected between the first electrode and the first bias power source. The second circuit is connected between the second electrode and the second bias power source. The second circuit has impedance higher than impedance of the first circuit at a common bias frequency of the first electrical bias and the second electrical bias.

[0005] The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, exemplary embodiments, and features described above, further aspects, exemplary embodiments, and features will become apparent by reference to the drawings and the following detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 schematically illustrates a plasma processing apparatus according to an exemplary embodiment.

[0007] FIG. 2 illustrates a configuration in a chamber of the plasma processing apparatus according to an exemplary embodiment.

[0008] FIG. 3 illustrates a first bias power source, a damping circuit, a first circuit, and a filter in a plasma processing apparatus according to an exemplary embodiment.

[0009] FIG. 4 illustrates a second bias power source, a damping circuit, a second circuit, and a filter in a plasma processing apparatus according to an exemplary embodiment.

[0010] FIG. 5A is a diagram showing simulation results of voltage waveforms in the plasma processing apparatus shown in FIG. 1, and FIG. 5B is a diagram showing simulation results of voltage waveforms in a plasma processing apparatus of a comparative example.

[0011] FIG. 6 illustrates a first bias power source, a damping circuit, a first circuit, and a filter in a plasma processing apparatus according to an exemplary embodiment.

[0012] FIG. 7 illustrates a second bias power source, a damping circuit, a second circuit, and a filter in a plasma processing apparatus according to an exemplary embodiment.

[0013] FIG. 8 schematically illustrates a plasma processing apparatus according to another exemplary embodiment.

[0014] FIG. 9 schematically illustrates a plasma processing apparatus according to still another exemplary embodiment.

DETAILED DESCRIPTION

[0015] Hereinafter, various exemplary embodiments will be described.

[0016] In an exemplary embodiment, a plasma processing apparatus is provided. The plasma processing apparatus includes a chamber, a substrate support, a first bias power source, a second bias power source, a first circuit, and a second circuit. The substrate support has a first region, a second region, a first electrode, and a second electrode. The first region is configured to support a substrate. The second region is configured to support an edge ring. The first electrode is provided in the first region. The second electrode is provided in the second region and separated from the first electrode. The first bias power source is configured to generate a first electrical bias and electrically connected to the first electrode. The second bias power source is configured to generate a second electrical bias and electrically connected to the second electrode. The first circuit is connected between the first electrode and the first bias power source. The second circuit is connected between the second electrode and the second bias power source. The second circuit has impedance higher than impedance of the first circuit at a common bias frequency of the first electrical bias and the second electrical bias.

[0017] In the plasma processing apparatus of the above embodiment, the first electrode and the substrate form a first capacitor element. Further, the second electrode and the edge ring form a second capacitor element. An area of the edge ring is generally smaller than an area of the substrate. Therefore, the capacitance of the second capacitor element is lower than the capacitance of the first capacitor element. Therefore, when an electric current that is supplied to the first capacitor element and an electric current that is supplied to the second capacitor element are the same as each other,

the voltage waveform of the edge ring changes at higher speed than the voltage waveform of the substrate. In the above embodiment, the first circuit is provided between the first electrode and the first bias power source, and the second circuit is provided between the second electrode and the second bias power source. At the bias frequency, the impedance of the second circuit is higher than the impedance of the first circuit. Therefore, the difference between the voltage waveform of the substrate and the voltage waveform of the edge ring is reduced.

[0018] In an exemplary embodiment, the impedance of the first circuit and the impedance of the second circuit are set such that a ratio between an electric current supplied to the substrate and an electric current supplied to the edge ring may be equal to a ratio between an area of the substrate and an area of the edge ring.

[0019] In an exemplary embodiment, the plasma processing may further include a controller. The controller may be configured to control the second bias power source to increase a setting level of the second electrical bias according to a decrease in a thickness of the edge ring and controls the second circuit to reduce the impedance of the second circuit according to the decrease in the thickness of the edge ring.

[0020] In an exemplary embodiment, each of the first electrical bias and the second electrical bias may be a pulse wave that is periodically generated at a cycle that is defined by the bias frequency. The pulse wave includes a pulse of a negative voltage. The pulse of the negative voltage may be a pulse of a negative direct-current voltage.

[0021] In an exemplary embodiment, each of the first electrical bias and the second electrical bias may be radio frequency power having the bias frequency.

[0022] In an exemplary embodiment, the first circuit may have a first resistor and a first capacitor. The first resistor is connected between the first electrode and the first bias power source. The first capacitor is connected between a node on an electrical path connecting the first resistor with the first electrode and a ground. The second circuit may have a second resistor and a second capacitor. The second resistor is connected between the second electrode and the second bias power source. The second capacitor is connected between a node on an electrical path connecting the second resistor with the second electrode and a ground. At least one of the second resistor or the second capacitor may be variable.

[0023] In an exemplary embodiment, the first circuit may have a first inductor and a first capacitor. The first inductor is connected between the first electrode and the first bias power source. The first capacitor is connected between a node on an electrical path connecting the first inductor with the first electrode and a ground. The second circuit may have a second inductor and a second capacitor. The second inductor is connected between the second electrode and the second bias power source. The second capacitor is connected between a node on an electrical path connecting the second inductor with the second electrode and a ground. At least one of the second inductor or the second capacitor may be variable.

[0024] Hereinafter, various exemplary embodiments will be described in detail with reference to the drawings. In the drawings, the same or equivalent portions are denoted by the same reference symbols.

[0025] FIG. 1 schematically illustrates a plasma processing apparatus according to an exemplary embodiment. A plasma processing apparatus 1 shown in FIG. 1 is provided with a chamber 10. FIG. 2 illustrates the configuration in the chamber of the plasma processing apparatus according to an exemplary embodiment. As shown in FIG. 2, the plasma processing apparatus 1 may be a capacitively coupled plasma processing apparatus.

[0026] The chamber 10 provides an internal space 10s therein. The central axis of the internal space 10s is an axis AX which extends in the vertical direction. In an embodiment, the chamber 10 includes a chamber body 12. The chamber body 12 has a substantially cylindrical shape. The internal space 10s is provided in the chamber body 12. The chamber body 12 is formed of, for example, aluminum. The chamber body 12 is electrically grounded. A film having plasma resistance is formed on the inner wall surface of the chamber body 12, that is, the wall surface defining the internal space 10s. This film may be a ceramic film such as a film formed by anodization or a film formed of yttrium oxide.

[0027] A passage 12p is formed in a side wall of the chamber body 12. A substrate W passes through the passage 12p when it is transferred between the internal space 10s and the outside of the chamber 10. A gate valve 12g is provided along the side wall of the chamber body 12 for opening and closing of the passage 12p.

[0028] The plasma processing apparatus 1 is further provided with a substrate support 16. The substrate support 16 is configured to support the substrate W placed thereon in the chamber 10. The substrate W has a substantially disk shape. The substrate support 16 is supported by a support 17. The support 17 extends upward from a bottom portion of the chamber body 12. The support 17 has a substantially cylindrical shape. The support 17 is formed of an insulating material such as quartz.

[0029] The substrate support 16 has a lower electrode 18 and an electrostatic chuck 20. The lower electrode 18 and the electrostatic chuck 20 are provided in the chamber 10. The lower electrode 18 is formed of a conductive material such as aluminum and has a substantially disk shape.

[0030] A flow path 18f is formed in the lower electrode 18. The flow path 18f is a flow path for a heat exchange medium. As the heat exchange medium, for example, a liquid refrigerant is used. A supply device for the heat exchange medium (for example, a chiller unit) is connected to the flow path 18f. The supply device is provided outside the chamber 10. The heat exchange medium is supplied from the supply device to the flow path 18f through a pipe 23a. The heat exchange medium supplied to the flow path 18f is returned to the supply device through a pipe 23b.

[0031] The electrostatic chuck 20 is provided on the lower electrode 18. As shown in FIG. 1, the electrostatic chuck 20 has a dielectric portion 20d and an electrode 21a. The electrostatic chuck 20 may further have an electrode 22a and an electrode 22b. When the substrate W is processed in the internal space 10s, the substrate W is placed on the electrostatic chuck 20 and is held by the electrostatic chuck 20. Further, an edge ring ER is mounted on the substrate support 16. The edge ring ER is a plate having a substantially ring shape. The edge ring ER has electrical conductivity. The edge ring ER is formed of, for example, silicon or silicon carbide. As shown in FIG. 2, the edge ring ER is mounted on the substrate support 16 such that the central axis thereof

coincides with the axis AX. The substrate W accommodated in the chamber 10 is disposed on the electrostatic chuck 20 and in a region surrounded by the edge ring ER.

[0032] The plasma processing apparatus 1 may be further provided with a gas line 25. The gas line 25 supplies a heat transfer gas, for example, a He gas, from a gas supply mechanism to a gap between the upper surface of the electrostatic chuck 20 (a first region to be described later) and the rear surface (lower surface) of the substrate W.

[0033] The plasma processing apparatus 1 may be further provided with an outer peripheral portion 28 and an outer peripheral portion 29. The outer peripheral portion 28 extends upward from the bottom portion of the chamber body 12. The outer peripheral portion 28 has a substantially cylindrical shape and extends along the outer periphery of the support 17. The outer peripheral portion 28 is formed of a conductive material. The outer peripheral portion 28 is electrically grounded. A film having plasma resistance is formed on the surface of the outer peripheral portion 28. This film may be a ceramic film such as a film formed by anodization or a film formed of yttrium oxide.

[0034] The outer peripheral portion 29 is provided on the outer peripheral portion 28. The outer peripheral portion 29 is formed of a material having insulation properties. The outer peripheral portion 29 is formed of ceramic such as quartz, for example. The outer peripheral portion 29 has a substantially cylindrical shape. The outer peripheral portion 29 extends along the outer peripheries of the lower electrode 18 and the electrostatic chuck 20.

[0035] The plasma processing apparatus 1 is further provided with an upper electrode 30. The upper electrode 30 is provided above the substrate support 16. The upper electrode 30 closes an upper opening of the chamber body 12 together with a member 32. The member 32 has insulation properties. The upper electrode 30 is supported on an upper portion of the chamber body 12 through the member 32.

[0036] The upper electrode 30 includes a ceiling plate 34 and a support 36. The lower surface of the ceiling plate 34 defines the internal space 10s. A plurality of gas discharge holes 34a are formed in the ceiling plate 34. Each of the plurality of gas discharge holes 34a penetrates the ceiling plate 34 in a plate thickness direction (the vertical direction). The ceiling plate 34 is formed of, for example, silicon. Alternatively, the ceiling plate 34 may have a structure in which a plasma-resistant film is provided on the surface of a member made of aluminum. This film may be a ceramic film such as a film formed by anodization or a film formed of yttrium oxide.

[0037] The support 36 detachably supports the ceiling plate 34. The support 36 is formed of a conductive material such as aluminum, for example. A gas diffusion chamber 36a is provided in the interior of the support 36. A plurality of gas holes 36b extend downward from the gas diffusion chamber 36a. The plurality of gas holes 36b communicate with the plurality of gas discharge holes 34a, respectively. A gas introduction port 36c is formed in the support 36. The gas introduction port 36c is connected to the gas diffusion chamber 36a. A gas supply pipe 38 is connected to the gas introduction port 36c.

[0038] A gas source group 40 is connected to the gas supply pipe 38 through a valve group 41, a flow rate controller group 42, and a valve group 43. The gas source group 40, the valve group 41, the flow rate controller group 42, and the valve group 43 configure a gas supply unit. The

gas source group 40 includes a plurality of gas sources. Each of the valve group 41 and the valve group 43 includes a plurality of valves (for example, on-off valves). The flow rate controller group 42 includes a plurality of flow rate controllers. Each of the plurality of flow rate controllers of the flow rate controller group 42 is a mass flow controller or a pressure control type flow rate controller. Each of the plurality of gas sources of the gas source group 40 is connected to the gas supply pipe 38 through a corresponding valve of the valve group 41, a corresponding flow rate controller of the flow rate controller group 42, and a corresponding valve of the valve group 43. The plasma processing apparatus 1 can supply gases from one or more gas sources selected from the plurality of gas sources of the gas source group 40 to the internal space 10s at individually adjusted flow rates.

[0039] A baffle plate 48 is provided between the outer peripheral portion 28 and the side wall of the chamber body 12. The baffle plate 48 may be configured, for example, by coating a member made of aluminum with ceramic such as yttrium oxide. A number of through-holes are formed in the baffle plate 48. An exhaust pipe 52 is connected to the bottom portion of the chamber body 12 below the baffle plate 48. An exhaust device 50 is connected to the exhaust pipe 52. The exhaust device 50 has a pressure controller such as an automatic pressure control valve, and a vacuum pump such as a turbo molecular pump, and can reduce the pressure in the internal space 10s.

[0040] Hereinafter, the substrate support 16 will be described in detail. As described above, the substrate support 16 has the lower electrode 18 and the electrostatic chuck 20. As shown in FIG. 1, the plasma processing apparatus 1 has a radio frequency power source 57. The radio frequency power source 57 is connected to the lower electrode 18 through a matcher 58. The radio frequency power source 57 is a power source that generates radio frequency power for plasma generation. The radio frequency power generated by the radio frequency power source 57 has a frequency within the range of 27 to 100 MHz, for example, a frequency of 40 MHz or 60 MHz. The matcher 58 has a matching circuit for matching the impedance on the load side (the lower electrode 18 side) of the radio frequency power source 57 with the output impedance of the radio frequency power source 57. The radio frequency power source 57 may not be electrically connected to the lower electrode 18, and may be connected to the upper electrode 30 through the matcher 58.

[0041] In the plasma processing apparatus 1, a radio frequency electric field is generated in the chamber 10 by the radio frequency power from the radio frequency power source 57. The gas in the chamber 10 is excited by the generated radio frequency electric field. As a result, plasma is generated in the chamber 10. The substrate W is processed with chemical species such as ions and/or radicals from the generated plasma.

[0042] The substrate support 16 has a first region 21 and a second region 22. The first region 21 is a central region of the substrate support 16. The first region 21 includes the central region of the electrostatic chuck 20 and the central region of the lower electrode 18. The second region 22 extends in a circumferential direction on the outside in a radial direction with respect to the first region 21. The second region 22 includes a peripheral edge region of the electrostatic chuck 20 and a peripheral edge region of the lower electrode 18. In the plasma processing apparatus 1, the

first region **21** and the second region **22** are configured from a single electrostatic chuck and are integrated with each other. In FIG. 1, the boundary between the first region **21** and the second region **22** is indicated by a broken line. In another embodiment, the first region **21** and the second region **22** may be configured from individual electrostatic chucks.

[0043] The first region **21** is configured to support the substrate **W** placed thereon (that is, on the upper surface thereof). The first region **21** is a region having a disk shape. The central axis of the first region **21** substantially coincides with the axis **AX**. The first region **21** shares the dielectric portion **20d** with the second region **22**. The dielectric portion **20d** is formed of a dielectric such as aluminum nitride or aluminum oxide. The dielectric portion **20d** has a substantially disk shape. In an embodiment, the thickness of the dielectric portion **20d** in the second region **22** is smaller than the thickness of the dielectric portion **20d** in the first region **21**. The position in the vertical direction of the upper surface of the dielectric portion **20d** in the second region **22** may be lower than the position in the vertical direction of the upper surface of the dielectric portion **20d** in the first region **21**.

[0044] The first region **21** has the electrode **21a** (chuck electrode). The electrode **21a** is an electrode having a film shape and is provided in the dielectric portion **20d** in the first region **21**. A direct-current power source **55** is connected to the electrode **21a** through a switch **56**. When a direct-current voltage from the direct-current power source **55** is applied to the electrode **21a**, an electrostatic attraction force is generated between the first region **21** and the substrate **W**. Due to the generated electrostatic attraction force, the substrate **W** is attracted to the first region **21** and held by the first region **21**.

[0045] The first region **21** further has a first electrode **21c**. The first electrode **21c** is an electrode having a film shape and is provided in the dielectric portion **20d** in the first region **21**. The electrode **21a** may extend closer to the upper surface of the first region **21** than the first electrode **21c** in the vertical direction.

[0046] The plasma processing apparatus **1** is further provided with a first bias power source **61**. The first bias power source **61** is electrically connected to the first electrode **21c** through a first circuit **63**. The first bias power source **61** generates a first electrical bias. The first electrical bias is applied to the first electrode **21c**. In an embodiment, the first electrical bias is a pulse wave that includes a pulse of a negative direct-current voltage and is periodically generated at a cycle that is defined by a bias frequency. The bias frequency may be a frequency in the range of 200 kHz to 13.56 MHz. The voltage level of the pulse wave may have a voltage value of 0 V or higher in a period other than a period in which the pulse of the negative direct-current voltage continues in the cycle, and the pulse wave may be, for example, a pulse wave having a positive or negative voltage value. Alternatively, the voltage of the pulse wave may have an absolute value lower than the absolute value of the voltage of the pulse in a period other than the period in which the pulse of the negative direct-current voltage continues in the cycle. The voltage level of the pulse may be temporally change within the cycle, and the pulse may be a pulse voltage such as a triangular wave or an impulse.

[0047] FIG. 3 illustrates a first bias power source, a damping circuit, a first circuit, and a filter in a plasma processing apparatus according to an exemplary embodiment. As shown in FIGS. 1 and 3, the plasma processing

apparatus **1** may be further provided with a damping circuit **62** and a filter **64**. The damping circuit **62** may be connected between the first bias power source **61** and the first circuit **63**. The filter **64** may be connected between the first circuit **63** and the first electrode **21c**. The first bias power source **61** may be connected to the first circuit **63** without going through the damping circuit **62**. In this case, the plasma processing apparatus **1** may not be provided with the damping circuit **62**.

[0048] As shown in FIG. 3, in an embodiment, the first bias power source **61** includes a variable direct-current power source **61p**, a switch **61a**, and a switch **61b**. The variable direct-current power source **61p** is a direct-current power source that generates a negative direct-current voltage. The level of the direct-current voltage that is generated by the variable direct-current power source **61p** is variable. The variable direct-current power source **61p** is connected to an output **61o** through the switch **61a**. The output **61o** is connected to the ground through the switch **61b**. The switch **61a** and the switch **61b** can be controlled by a controller **MC** (described later). In a case where the switch **61a** is in a conduction state and the switch **61b** is in a non-conduction state, a negative direct-current voltage is output from the output **61o**. In a case where the switch **61a** is in the non-conduction state and the switch **61b** is in the conduction state, the voltage level of the output **61o** becomes 0 V. A pulse wave, which is the first electrical bias, can be obtained by controlling the conduction state of each of the switch **61a** and the switch **61b**.

[0049] The damping circuit **62** is connected between the output **61o** of the first bias power source **61** and the first circuit **63**. In an embodiment, the damping circuit **62** has a resistor **62r** and a capacitor **62c**. One end of the resistor **62r** is connected to the output **61o** of the first bias power source **61**. One end of the capacitor **62c** is connected to a node **62n** on an electrical path connecting the other end of the resistor **62r** with the first circuit **63**. The other end of the capacitor **62c** is grounded.

[0050] The impedance of the first circuit **63** may be variable. The first circuit **63** has one or more variable circuit elements. Each of the one or more variable circuit elements has a variable element parameter. In an embodiment, the first circuit **63** has a first variable resistor **63r** and a first variable capacitor **63c** as the one or more variable circuit elements. In the first circuit **63**, the variable element parameters are the resistance value of the first variable resistor **63r** and the capacitance of the first variable capacitor **63c**. One end of the first variable resistor **63r** is connected to the output **61o** of the first bias power source **61** through the damping circuit **62**. One end of the first variable capacitor **63c** is connected to a node **63n** on an electrical path connecting the other end of the first variable resistor **63r** with the first electrode **21c**. The other end of the first variable capacitor **63c** is grounded. The impedance of the first circuit **63** is set by the controller **MC**. The impedance of the first circuit **63** is controlled by setting the variable element parameter of each of one or more variable circuit elements of the first circuit **63**, for example, the resistance value of the first variable resistor **63r** and the capacitance of the first variable capacitor **63c**, by the controller **MC**. The impedance of the first circuit **63** may be constant rather than variable. That is, a fixed resistor may be used instead of the first variable resistor **63r**, and a fixed capacitor may be used instead of the first variable capacitor **63c**.

[0051] The filter 64 is connected between the node 63_n and the first electrode 21_c. The filter 64 is an electric filter configured to block or attenuate the radio frequency power from the radio frequency power source 57. The filter 64 includes, for example, an inductor connected between the node 63_n and the first electrode 21_c.

[0052] As shown in FIG. 1, the second region 22 extends to surround the first region 21. The second region 22 is a substantially annular region. The central axis of the second region 22 substantially coincides with the axis AX. The second region 22 is configured to support the edge ring ER placed thereon (that is, on the upper surface thereof). The second region 22 shares the dielectric portion 20_d with the first region 21.

[0053] In an embodiment, the second region 22 may hold the edge ring ER by an electrostatic attraction force. In this embodiment, the second region 22 may have one or more electrodes (chuck electrodes). In the embodiment shown in FIG. 1, the second region 22 has a pair of electrodes, that is, the electrode 22_a and the electrode 22_b. The electrode 22_a and the electrode 22_b are provided in the dielectric portion 20_d in the second region 22. The electrode 22_a and the electrode 22_b configure a bipolar electrode. Each of the electrode 22_a and the electrode 22_b is an electrode having a film shape. The electrode 22_a and the electrode 22_b may extend at substantially the same height position in the vertical direction.

[0054] A direct-current power source 71 is connected to the electrode 22_a through a switch 72 and a filter 73. The filter 73 is an electric filter configured to block or attenuate the radio frequency power and the first and second electrical biases. A direct-current power source 74 is connected to the electrode 22_b through a switch 75 and a filter 76. The filter 76 is an electric filter configured to block or reduce the radio frequency power and the first and second electrical biases.

[0055] The direct-current power source 71 and the direct-current power source 74 apply direct-current voltages to the electrodes 22_a and 22_b, respectively, in order to generate an electrostatic attraction force that attracts the edge ring ER to the second region 22. The setting potential of each of the electrodes 22_a and 22_b may be any of positive potential, negative potential, and 0 V. For example, the potential of the electrode 22_a may be set to positive potential, and the potential of the electrode 22_b may be set to negative potential. Further, the potential difference between the electrode 22_a and the electrode 22_b may be formed by using a single direct-current power source instead of the two direct-current power sources.

[0056] When a direct-current voltage is applied to the electrode 22_a and the electrode 22_b, an electrostatic attraction force is generated between the second region 22 and the edge ring ER. The edge ring ER is attracted to the second region 22 by the generated electrostatic attraction force and held by the second region 22.

[0057] The second region 22 further has a second electrode 22_c. The second electrode 22_c is an electrode having a film shape. The second electrode 22_c is provided in the dielectric portion 20_d in the second region 22. The second electrode 22_c is separated from the first electrode 21_c. The electrode 22_a and the electrode 22_b may extend closer to the upper surface of the second region 22 than the second electrode 22_c in the vertical direction. The second electrode 22_c may be disposed outside the second region 22. For

example, the second electrode 22_c may be provided below the edge ring ER and in the outer peripheral portion 29.

[0058] The plasma processing apparatus 1 is further provided with a second bias power source 81. The second bias power source 81 is electrically connected to the second electrode 22_c through a second circuit 83. The second bias power source 81 generates a second electrical bias. The second electrical bias is applied to the second electrode 22_c. In an embodiment, the second electrical bias is a pulse wave that includes a pulse of a negative direct-current voltage and is periodically generated at a cycle that is defined by the bias frequency. The bias frequency of the second electrical bias is the same as the bias frequency of the first electrical bias. The voltage level of the pulse wave may have a voltage value of 0 V or higher in a period other than a period in which the pulse of the negative direct-current voltage continues in the cycle, and the pulse wave may be, for example, a pulse wave having a positive or negative voltage value. Alternatively, the voltage of the pulse wave may have an absolute value lower than the absolute value of the voltage of the pulse in a period other than the period in which the pulse of the negative direct-current voltage continues in the cycle. The voltage level of the pulse may temporally change within the cycle, and the pulse may be a pulse voltage such as a triangular wave or an impulse.

[0059] FIG. 4 illustrates a second bias power source, a damping circuit, a second circuit, and a filter in a plasma processing apparatus according to an exemplary embodiment. As shown in FIGS. 1 and 4, the plasma processing apparatus 1 may be further provided with a damping circuit 82 and a filter 84. The damping circuit 82 may be connected between the second bias power source 81 and the second circuit 83. The filter 84 may be connected between the second circuit 83 and the second electrode 22_c. The second bias power source 81 may be connected to the second circuit 83 without going through the damping circuit 82. In this case, the plasma processing apparatus 1 may not be provided with the damping circuit 82.

[0060] As shown in FIG. 4, in an embodiment, the second bias power source 81 includes a variable direct-current power source 81_p, a switch 81_a, and a switch 81_b. The variable direct-current power source 81_p is a direct-current power source that generates a negative direct-current voltage. The level of the direct-current voltage that is generated by the variable direct-current power source 81_p is variable. The variable direct-current power source 81_p is connected to an output 81_o through the switch 81_a. The output 81_o is connected to the ground through the switch 81_b. The switch 81_a and the switch 81_b can be controlled by the controller MC (described later). In a case where the switch 81_a is in a conduction state and the switch 81_b is in a non-conduction state, a negative direct-current voltage is output from the output 81_o. In a case where the switch 81_a is in the non-conduction state and the switch 81_b is in the conduction state, the voltage level of the output 81_o becomes 0 V. A pulse wave, which is the second electrical bias, can be obtained by controlling the conduction state of each of the switch 81_a and the switch 81_b.

[0061] The damping circuit 82 is connected between the output 81_o of the second bias power source 81 and the second circuit 83. In an embodiment, the damping circuit 82 has a resistor 82_r and a capacitor 82_c. One end of the resistor 82_r is connected to the output 81_o of the second bias power source 81. One end of the capacitor 82_c is connected to a

node **82n** on an electrical path connecting the other end of the resistor **82r** with the second circuit **83**. The other end of the capacitor **82c** is grounded.

[0062] The impedance of the second circuit **83** may be variable. The second circuit **83** has one or more variable circuit elements. Each of the one or more variable circuit elements has a variable element parameter. In an embodiment, the second circuit **83** has a second variable resistor **83r** and a second variable capacitor **83c** as the one or more variable circuit elements. In the second circuit **83**, the variable element parameters are the resistance value of the second variable resistor **83r** and the capacitance of the second variable capacitor **83c**. One end of the second variable resistor **83r** is connected to the output **81o** of the second bias power source **81** through the damping circuit **82**. One end of the second variable capacitor **83c** is connected to a node **83n** on an electrical path connecting the other end of the second variable resistor **83r** with the second electrode **22c**. The other end of the second variable capacitor **83c** is grounded. The second circuit **83** has impedance higher than the impedance of the first circuit **63** at a common bias frequency of the first electrical bias and the second electrical bias. In an embodiment, the impedance of the first circuit **63** and the impedance of the second circuit **83** are set such that the ratio between an electric current that is supplied to the substrate **W** and an electric current that is supplied to the edge ring **ER** is equal to the ratio between an area of the substrate **W** and an area of the edge ring **ER**. The impedance of the second circuit **83** is set by the controller **MC**. The impedance of the second circuit **83** is controlled by setting the variable element parameter of each of one or more variable circuit elements of the second circuit **83**, for example, the resistance value of the second variable resistor **83r** and the capacitance of the second variable capacitor **83c**, by the controller **MC**. The impedance of the second circuit **83** may be constant rather than variable. That is, a fixed resistor may be used instead of the second variable resistor **83r**, and a fixed capacitor may be used instead of the second variable capacitor **83c**.

[0063] The filter **84** is connected between the node **83n** and the second electrode **22c**. The filter **84** is an electric filter configured to block or attenuate the radio frequency power from the radio frequency power source **57**. The filter **84** includes, for example, an inductor connected between the node **83n** and the second electrode **22c**.

[0064] The second region **22** may further have a gas line **22g**. The gas line **22g** is a gas line provided for supplying a heat transfer gas, for example, a He gas, to the gap between the second region **22** and the edge ring **ER**. The gas line **22g** is connected to a gas supply mechanism **86** which is a heat transfer gas source.

[0065] In an embodiment, as shown in FIG. 2, the plasma processing apparatus **1** is further provided with the controller **MC**. The controller **MC** is a computer which includes a processor, a storage device, an input device, a display device, and the like, and controls each part of the plasma processing apparatus **1**. Specifically, the controller **MC** executes a control program stored in the storage device, and controls each part of the plasma processing apparatus **1**, based on recipe data stored in the storage device. The process designated by the recipe data is performed in the plasma processing apparatus **1** under the control by the controller **MC**.

[0066] Here, the edge ring **ER** wears by being exposed to plasma, so that the thickness thereof decreases. In a case where the thickness of the edge ring **ER** becomes smaller than the initial thickness thereof, the upper end of a sheath (plasma sheath) is inclined in the vicinity of an edge of the substrate **W**. Therefore, in a case where the thickness of the edge ring **ER** becomes smaller than the initial thickness thereof, an incident direction of ions with respect to the edge of the substrate **W** is inclined with respect to the vertical direction. In an embodiment, the controller **MC** may control the second bias power source **81** to increase the setting level of the second electrical bias according to a decrease in the thickness of the edge ring **ER**. In a case where the second electrical bias is the pulse wave described above, the setting level of the second electrical bias is the absolute value of the voltage of the pulse in the pulse wave. When the setting level of the second electrical bias is increased, the thickness of the sheath increases above the edge ring **ER**, so that the inclination of the incident direction of the ions with respect to the edge of the substrate **W** can be corrected.

[0067] The controller **MC** may specify the setting level of the second electrical bias corresponding to the thickness of the edge ring **ER** by using a function or a table stored in the storage device thereof. The thickness of the edge ring **ER** may be measured optically or electrically, or may be estimated from a time in which the edge ring **ER** is exposed to plasma.

[0068] Further, the controller **MC** may control the variable element parameter of each of the one or more variable circuit elements of the second circuit **83** to reduce the impedance of the second circuit **83** according to the decrease in the thickness of the edge ring **ER**. The impedance of the second circuit **83** is reduced according to an increase in the setting level of the second electrical bias, whereby an increase in a time length that is required to reach a peak level from a base level in the voltage waveform of the edge ring **ER** is suppressed.

[0069] In an embodiment, the controller **MC** may specify the variable element parameter of each of the one or more variable circuit elements of the second circuit **83** corresponding to the setting level of the second electrical bias by using a function or a table stored in the storage device thereof. The variable element parameter of each of the one or more variable circuit elements of the second circuit **83** corresponding to the thickness of the edge ring **ER** may be directly associated with the thickness of the edge ring **ER**.

[0070] In the plasma processing apparatus **1**, the first electrode **21c** and the substrate **W** form a first capacitor element. Further, the second electrode **22c** and the edge ring **ER** form a second capacitor element. The region of the edge ring **ER** is smaller than the region of the substrate **W**. Therefore, the capacitance of the second capacitor element is lower than the capacitance of the first capacitor element. Therefore, when an electric current that is supplied to the first capacitor element and an electric current that is supplied to the second capacitor element are the same as each other, the voltage waveform of the edge ring **ER** changes at a higher speed than the voltage waveform of the substrate **W**. In the plasma processing apparatus **1**, the first circuit **63** is provided between the first electrode **21c** and the first bias power source **61**, and the second circuit **83** is provided between the second electrode **22c** and the second bias power source **81**. At the bias frequency, the impedance of the second circuit **83** is set to impedance higher than the

impedance of the first circuit **63**. Therefore, according to the plasma processing apparatus **1**, the difference between the voltage waveform of the substrate **W** and the voltage waveform of the edge ring **ER** is reduced.

[0071] Hereinafter, FIGS. **5A** and **5B** will be referred to. FIG. **5A** is a diagram showing simulation results of voltage waveforms in the plasma processing apparatus shown in FIG. **1**. FIG. **5B** is a diagram showing simulation results of voltage waveforms in a plasma processing apparatus of a comparative example. The plasma processing apparatus of the comparative example is a plasma processing apparatus in which the first circuit **63** and the second circuit **83** are removed from the plasma processing apparatus **1**. In each of FIGS. **5A** and **5B**, the horizontal axis represents time and the vertical axis represents voltage. In each of FIGS. **5A** and **5B**, the waveform of the output voltage (the second electrical bias) of the second bias power source **81**, the voltage waveform of the edge ring **ER**, and the voltage waveform of the substrate **W** are respectively indicated by a dashed-dotted line, a solid line, and a broken line. As shown in FIG. **5B**, in the plasma processing apparatus of the comparative example which does not have the first circuit **63** and the second circuit **83**, a difference occurs between the voltage waveform of the substrate **W** and the voltage waveform of the edge ring **ER**. On the other hand, as shown in FIG. **5A**, in the plasma processing apparatus **1**, the difference between the voltage waveform of the substrate **W** and the voltage waveform of the edge ring **ER** is reduced.

[0072] Hereinafter, FIGS. **6** and **7** will be referred to. FIG. **6** illustrates a first bias power source, a damping circuit, a first circuit, and a filter in a plasma processing apparatus according to an exemplary embodiment. FIG. **7** illustrates a second bias power source, a damping circuit, a second circuit, and a filter in a plasma processing apparatus according to an exemplary embodiment. As shown in FIG. **6**, the first circuit **63** may have a first variable inductor **63i** instead of the first variable resistor **63r**. Further, as shown in FIG. **7**, the second circuit **83** may have a second variable inductor **83i** instead of the second variable resistor **83r**. Although the example in which the first circuit and the second circuit include the variable elements is shown, the first circuit and/or the second circuit may not include the variable element.

[0073] Hereinafter, FIG. **8** will be referred to. FIG. **8** schematically illustrates a plasma processing apparatus according to another exemplary embodiment. A plasma processing apparatus **1B** shown in FIG. **8** includes a radio frequency bias power source as the first bias power source **61**. The plasma processing apparatus **1B** includes a radio frequency bias power source as the second bias power source **81**. In the plasma processing apparatus **1B**, the first bias power source **61** is configured to generate radio frequency bias power having a bias frequency as the first electrical bias. The bias frequency is a frequency within the range of 200 kHz to 13.56 MHz, and is, for example, 400 kHz. In the plasma processing apparatus **1B**, the first bias power source **61** is connected to the first electrode **21c** through a matcher **65** and the first circuit **63**. The matcher **65** has a matching circuit for matching the impedance on the load side of the first bias power source **61** with the output impedance of the first bias power source **61**.

[0074] Further, in the plasma processing apparatus **1B**, the second bias power source **81** is configured to generate radio frequency bias power having a bias frequency as the second

electrical bias. The bias frequency of the radio frequency bias power that is generated by the second bias power source **81** is the same as the bias frequency of the radio frequency bias power that is generated by the first bias power source **61**. Further, in the plasma processing apparatus **1B**, the second bias power source **81** is connected to the second electrode **22c** through a matcher **85** and the second circuit **83**. The matcher **85** has a matching circuit for matching the impedance on the load side of the second bias power source **81** with the output impedance of the second bias power source **81**. In the plasma processing apparatus **1B**, the setting level of the second electrical bias that is controlled by the controller **MC** is a power level of the radio frequency bias power. Other configurations of the plasma processing apparatus **1B** may be the same as the corresponding configurations of the plasma processing apparatus **1**.

[0075] Hereinafter, FIG. **9** will be referred to. FIG. **9** schematically illustrates a plasma processing apparatus according to still another exemplary embodiment. In a plasma processing apparatus **1C** shown in FIG. **9**, the electrode **22a** and the electrode **22b** are used as the second electrode **22c**. The electrical path extending from the output of the second bias power source **81** is branched into two branch paths in a subsequent stage of the second circuit **83** (or the filter **84**), and the two branch paths are respectively connected to the electrode **22a** and the electrode **22b** through blocking capacitors **87a** and **87b**. Other configurations of the plasma processing apparatus **1C** may be the same as the corresponding configurations of the plasma processing apparatus **1**. Also in the plasma processing apparatus **1B**, similar to the plasma processing apparatus **1C**, the electrodes **22a** and **22b** may be used as the second electrode to which the second electrical bias is applied, and the second electrode **22c** separate from the electrodes **22a** and **22b** may be omitted.

[0076] While various exemplary embodiments have been described above, various additions, omissions, substitutions and changes may be made without being limited to the exemplary embodiments described above. Elements of the different embodiments may be combined to form another embodiment.

[0077] For example, in another embodiment, the plasma processing apparatus may be a capacitively coupled plasma processing apparatus different from the plasma processing apparatus **1**. In another embodiment, the plasma processing apparatus may be another type of plasma processing apparatus. The other type of plasma processing apparatus may be an inductively coupled plasma processing apparatus, an electron cyclotron resonance (ECR) plasma processing apparatus, or a plasma processing apparatus that generates plasma by using surface waves such as microwaves.

[0078] Further, the first electrode **21c** and the second electrode **22c** may not be provided in the dielectric portion **20d** of the electrostatic chuck **20**. Each of the first electrode **21c** and the second electrode **22c** may be provided in another dielectric portion provided between the electrostatic chuck **20** and the lower electrode **18**.

[0079] Further, each of the one or more variable circuit elements in each of the first circuit **63** and the second circuit **83** is not a single variable circuit element, but may be configured with an array of a plurality of fixed circuit elements and a plurality of switching elements respectively connected to the plurality of fixed circuit elements. In this

case, the number of fixed circuit elements that are connected in parallel is adjusted by controlling the plurality of switching elements.

[0080] From the foregoing description, it will be appreciated that various embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

What is claimed is:

1. A plasma processing apparatus comprising:
 - a chamber;
 - a substrate support provided in the chamber and having a first region configured to support a substrate, a second region configured to support an edge ring, a first electrode provided in the first region, and a second electrode provided in the second region and separated from the first electrode;
 - a first bias power source configured to generate a first electrical bias and electrically connected to the first electrode;
 - a second bias power source configured to generate a second electrical bias and electrically connected to the second electrode;
 - a first circuit connected between the first electrode and the first bias power source; and
 - a second circuit connected between the second electrode and the second bias power source and having impedance higher than impedance of the first circuit at a common bias frequency of the first electrical bias and the second electrical bias.
2. The plasma processing apparatus according to claim 1, wherein the impedance of the first circuit and the impedance of the second circuit are set such that a ratio between an electric current supplied to the substrate and an electric current supplied to the edge ring is equal to a ratio between an area of the substrate and an area of the edge ring.
3. The plasma processing apparatus according to claim 1, further comprising:
 - a controller configured to control the second bias power source and the second circuit,
 wherein the controller is configured to control the second bias power source to increase a setting level of the second electrical bias according to a decrease in a thickness of the edge ring and controls the second circuit to reduce the impedance of the second circuit according to the decrease in the thickness of the edge ring.
4. The plasma processing apparatus according to claim 2, further comprising:
 - a controller configured to control the second bias power source and the second circuit,
 wherein the controller is configured to control the second bias power source to increase a setting level of the second electrical bias according to a decrease in a thickness of the edge ring and controls the second circuit to reduce the impedance of the second circuit according to the decrease in the thickness of the edge ring.
5. The plasma processing apparatus according to claim 1, wherein each of the first electrical bias and the second electrical bias is a pulse wave that includes a pulse of a

negative voltage and is periodically generated at a cycle that is defined by the bias frequency.

6. The plasma processing apparatus according to claim 5, wherein the pulse of the negative voltage is a pulse of a negative direct-current voltage.

7. The plasma processing apparatus according to claim 1, wherein each of the first electrical bias and the second electrical bias is radio frequency power having the bias frequency.

8. The plasma processing apparatus according to claim 1, wherein

the first circuit has

- a first resistor connected between the first electrode and the first bias power source, and

- a first capacitor connected between a node on an electrical path connecting the first resistor with the first electrode and a ground, and

the second circuit has

- a second resistor connected between the second electrode and the second bias power source, and

- a second capacitor connected between a node on an electrical path connecting the second resistor with the second electrode and a ground.

9. The plasma processing apparatus according to claim 8, wherein at least one of the second resistor or the second capacitor is variable.

10. The plasma processing apparatus according to claim 5, wherein

the first circuit has

- a first resistor connected between the first electrode and the first bias power source, and

- a first capacitor connected between a node on an electrical path connecting the first resistor with the first electrode and a ground, and

the second circuit has

- a second resistor connected between the second electrode and the second bias power source, and

- a second capacitor connected between a node on an electrical path connecting the second resistor with the second electrode and a ground.

11. The plasma processing apparatus according to claim 10, wherein at least one of the second resistor or the second capacitor is variable.

12. The plasma processing apparatus according to claim 1, wherein

the first circuit has

- a first inductor connected between the first electrode and the first bias power source, and

- a first capacitor connected between a node on an electrical path connecting the first inductor with the first electrode and a ground, and

the second circuit has

- a second inductor connected between the second electrode and the second bias power source, and

- a second capacitor connected between a node on an electrical path connecting the second inductor with the second electrode and a ground.

13. The plasma processing apparatus according to claim 12, wherein at least one of the second inductor or the second capacitor is variable.

14. The plasma processing apparatus according to claim 5, wherein

the first circuit has

a first inductor connected between the first electrode and the first bias power source, and

a first capacitor connected between a node on an electrical path connecting the first inductor with the first electrode and a ground, and

the second circuit has

a second inductor connected between the second electrode and the second bias power source, and

a second capacitor connected between a node on an electrical path connecting the second inductor with the second electrode and a ground.

15. The plasma processing apparatus according to claim **14**, wherein at least one of the second inductor or the second capacitor is variable.

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